1. Consider the abstract data type listADT of list of integers defined in the lecture notes. Assume that the concrete implementation version 2.0 (i.e., linked list based) is used. Consider the following function Fn1.

```
listADT Fn1(int n, listADT L) {       if
(ListIsEmpty(L)) return Cons(n,
EmptyList());   if (n <= Head(L)) return
Cons(n, L);     return Cons(Head(L),
Fn1(n, Tail(L)));
}
```

   a) What is the computational complexity of function ListIsEmpty?
   b) What is the computational complexity of function EmptyList?
   c) What is the computational complexity of function Cons?
   d) What is the computational complexity of function Head?
   e) What is the computational complexity of function Tail?
   f) What is the computational complexity of function Fn1 if L is assumed to have been sorted into ascending order?
   g) What is the computational complexity of function Fn1 if L is not assumed to have been sorted?

a) O(1)
b) O(1)
c) O(1)
d) O(1)
e) O(1)
f) O(n)
g) O(n)